DIRECT/COMMUNITY BROADCAST PROJECTS USING SPACE SATELLITES

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The extended discussions of direct broadcasting by means of space satellites, which have for some time preoccupied international forums such as the United Nations and UNESCO, can be placed in useful perspective through a review of those direct/community broadcast projects which are currently in preparation in India, Canada, and Japan. The international debates are beset by abstractions such as "sovereignty" on the one side and "freedom of information" on the other, whose degree of relevancy is in fact set by technical production, distribution, and control realities. These realities constrain prospective broadcasters as well as prospective receivers. Indeed, the legalisms which are argued in political forums may, in good part, never be reached in the real world of broadcast operations. Furthermore, the technical prospects presented by space broadcasting suggest that problems of intrusion and interference may be substantially less difficult to handle than these same problems have been in the past half century of radio broadcasting, an entirely analogous activity.

A word should be said regarding the special concern which has been attached in international debate to the problems of TV broadcasting as distinguished from radio broadcasting. Since the two differ in principle only in that TV broadcasting adds a visual image to the audio element, it may be presumed that the fear of intrusive broadcasting (by one country direct to receivers in another) is greater in the case of TV than in the case of radio because of the greater audience impact which TV is presumed to have. We will assume for purposes of this discussion that this presumption is correct, as indeed it may be.

In the light of these few introductory observations, a brief review of the Indian, Canadian and Japanese experimental space broadcasting projects follows. During the review, some light will be shed on the broader issues of direct/community broadcasting by satellite.

I.

At the outset, it will be seen in all three projects that the governments concerned are initiating broadcast experiments for practical domestic reasons. In India, there is no nationwide diffusion system capable of reaching a majority of the population, which is substantially illiterate, thus placing a great premium on the visual values of TV. There are at this writing only two television broadcast stations in all India, although others are

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under construction. The population is relatively evenly distributed in over half a million villages throughout the country so that it would be extremely difficult to cover a large proportion of the population through conventional television broadcast facilities located in a few urban centers.

Moreover, the great social problems confronting the government of India, including population control and agricultural productivity, can be attacked only through education and communication directed to education. Thus, the advanced technology of direct broadcast satellites which, at a stroke, brings the entire subcontinent within reach of a single broadcasting point, becomes extraordinarily attractive to India. A direct satellite broadcasting service offers the prospect of a national diffusion system, not only for TV but also for telephone and other services, within a very short period of time and at a fraction of the cost required for conventional ground-based distribution systems.[1]

In Canada a very different problem confronts the government. More than 95 percent of Canada's population is located in roughly 10 percent of Canada's territory, that area south of a line of latitude crossing slightly north of Ottawa. In the remaining 90 percent of Canada's territory some 2 percent of the population is very thinly distributed. In terms of strict cost-benefit considerations, one might conclude that special communications provisions for the remote population group could not be justified. Nevertheless, the Canadian government places high value on extending to that population the benefits of educational, news, and entertainment programs. In addition, effective links into this region would serve important government and commercial purposes. Space broadcasting techniques offer an attractive means to reach thinly distributed populations, not only in Canada, but also in the hinterland of Brazil, Alaska, and other such areas. Such techniques make it possible to reach remote receivers with no surface link to them or between them at relatively low cost.[2]

In Japan the problems are again different. From a communications viewpoint, Japan presents special difficulties of fragmentation and shadow. Fragmentation of communications systems is occasioned by the multi-island character of the nation. In addition, the mountainous character of the country and the presence of tall urban structures present shadow problems interfering with reception of TV signals from ground-based diffusion points. Here again, the space satellite affords direct contact to receivers on the ground without regard to insular location or urban shadow.[3]

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[1]Indian Space Research Organization, Satellite Instructional Television Experiment: Television to Remote Areas Through Satellite (1974); Address by Yash Pal, Indian Experiment in Satellite Broadcasting—Some Experiments, United Nations Panel Meeting on Satellite Broadcasting Systems for Education, Feb. 26-Mar. 7, 1974. *See also* Hearings on S. 2955 Before the Senate Comm. on Aeronautical and Space Sciences, 93rd Cong., 2nd Sess., pt. 2 at 768 (1974).

[2]B. Blevis & N. Davies, Investigation of the Advanced Communications Satellites (report prepared for the Department of Communications, Communications Research Center, Ottawa, Canada); Communications Canada, Oct. 17-19, 1974 (text submitted to a workshop discussion at the Interdisciplinary Conference on the Development of the Mid-North, Chicoutimi, Quebec, Canada). *See also* Hearings, *supra* note 1, at 668.

[3]H. Chiba, Talking Paper on Communications Satellite and Broadcasting Satellite, January 17, 1974 (Mr. Chiba is Director General of the Research Coordination Bureau of the Japanese Science and Technology Agency, Tokyo).

A first point of some significance for international discussion may be made at this juncture. The three pioneering projects under review all reflect very real *domestic* needs and concerns. There has been no sign of planning for non-domestic, unilateral programming by the major powers or by smaller countries engaged in international conflicts, the countries most often in the minds of those who fear the advent of intrusive broadcasting programs offensive in terms of program value or propaganda character. Even if a country's interests were to extend beyond domestic purposes, it will be shown that such a country could hope for very little beyond *mutually-desired* broadcast contact with the populations of other countries.

II.

Let us now examine in more detail each of the projects planned by the three countries under consideration, beginning first with India. The Indian project stems from concepts of NASA's Office of International Affairs, pursued in parallel within the U.S. government and by Indian space interests. Preliminary exploration began in the mid-sixties and progressed to the point where a one-year study was undertaken in 1967 jointly by NASA and the Indian Space Research Organization (ISRO) to determine whether an experiment would have practical follow-on value for India. The conclusion was that India could anticipate that such an experiment would lead directly to a multi-purpose domestic space communications system with substantial cost and time savings over conventional diffusion systems. At the same time, India undertook a test of the real effectiveness of TV for instructional programs for villages. In this test, conventional TV receivers were placed in some 80 villages surrounding New Delhi and a conventional broadcast station beamed programs, principally of an agricultural content, to the receiving villages. The impact on farm productivity in the receiver-equipped villages was compared with productivity in control villages not so equipped, and the results were considered favorable.[4]

In 1969, an agreement was reached between the Indian Space Research Organization and NASA by which the ATS-F satellite, the first satellite capable of broadcasting directly into small receivers, would be contributed for one year to a large-scale experiment in community broadcasting to some 5,000 Indian villages. That experiment, named Satellite Instructional Television Experiment (SITE), has been under intensive preparation since 1969 and is scheduled now to begin in August 1975. Some 2,000 villages will receive Indian programs relayed through the U.S. satellite to essentially conventional TV receivers *augmented* by means of a parabolic antenna of 7 to 10 feet in diameter, a modulation converter, and a pre-amplifier. The remaining villages will utilize conventional receivers that obtain their broadcast signals from small, ground relay stations that receive signals from the satellite. (Economies can be realized where the density of receivers is such that sufficient conventional, *i.e.,* non-augmented, receivers can be installed to offset the added cost of a ground relay station.[5])

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[4]Indian Space Research Organization, *supra* note 1; Pal, *supra* note 1.

[5]*Id.*

The satellite to be used was developed by NASA for domestic programs and radio propagation experiments of wide-ranging character. It broadcasts with 80 watts of transmission power through a high-gain antenna. The satellite points its beam with an accuracy of 0.1 degree. For the India experiment it will broadcast at 860 MHz, a frequency acceptable for experimental purposes only and not to be used for operational space communications purposes. (The ITU has already designated other frequency bands for operational direct broadcasting.[6]) There will be one video channel available to India and two voice channels, permitting use of several languages through time-sharing. India's objectives are to conduct instructional programs, especially in the population control and agricultural productivity fields, and programs contributing to national coherence. The audience will consist largely of illiterate adult villagers who have generally not been reachable in any other way. *All of the programming will be developed in India by Indian agencies;* thus, no U.S. software will be involved. Local village participation in the programming and in feedback systems for criticizing and modifying the programming will be sought by Indian authorities.[7]

The respective contributions of the U.S. and India to the project are essentially these: While there are some incremental costs for the United States in supporting SITE, these represent very small fractions of the cost of the total U.S. domestic program planned with the ATS-F satellite. The real contribution of the United States is one year of the satellite's life to India for the SITE program. India, on the other hand, is responsible for providing a ground transmitter, for the design, manufacture, distribution, installation, and maintenance of the 5,000 receivers required, for preparation of the instructional programs to be broadcast, and for social evaluation of the program. India, of course, will also have to meet other logistics requirements, such as making available to villagers those agricultural productivity items (*e.g.*, fertilizers) and population control devices whose presence is implied by the programs. It is estimated that India's total program cost will be approximately $20 million.

The India experiment is, of course, of prime significance for developing countries—those which have not been able to reach large segments of their population, those which have overriding social problems which might be ameliorated through communication and education, and particularly those where visual techniques could help to by-pass prevalent illiteracy.

No serious question of intrusive broadcasting is presented by the Indian project. The programs will be receivable only through instruments designed and built in India especially for the purpose. Neighboring countries could receive the programs only if, *wishing to do so,* they procure receivers from India or deliberately construct receivers for that purpose. However, since the down-link SITE frequency of 860 MHz could interfere

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[6]For designated frequency bands for operational direct broadcasting, see Partial Revision of the Radio Regulations, Geneva, 1971 and Final Protocol: Space Telecommunications, July 17, 1971, [1972] 23 U.S.T. 1527, T.I.A.S. No. 7435 (effective Jan. 1, 1973).

[7] NASA, The ATS-F Data Handbook (rev. May 1974).

with terrestrial services in Europe, the satellite will be positioned for the India experiment over the east coast of Africa; its antenna will be pointed eastward toward India and away from Europe, thus avoiding interference with European surface broadcasts.

III.

Canada will be the first nation to utilize radio frequencies in the 12-14 GHz bands designated by the ITU for space broadcasting purposes. There are very great advantages in these frequency ranges. Terrestrial use of these frequencies is relatively limited and so no major interference problems are anticipated. Consequently, there are no power flux problems, *i.e.*, no limits on the power which can be used in the satellite to transmit to ground receivers. This situation makes it possible to concentrate the power requirements of the system in the satellite and to use small and relatively inexpensive receivers on the ground. At these broadcast frequencies, such receivers can be used anywhere, even in urban centers to get around the problem of urban shadow.[8]

The Canadian project will be performed as a cooperative U.S./Canadian space program. Canada initiated the proposal and will build the required satellite at a cost of roughly $40 million. The United States (NASA) will contribute the principal power source for the experiment in the satellite, a 200-watt traveling wave tube, and, as a further contribution, will launch the satellite in late 1975. (The European Space Research Organization will also participate and will contribute a 20-watt traveling wave tube plus other subsystems.) The satellite will maintain its station above the equator with an accuracy of plus or minus 0.2 degree and with comparable pointing accuracy. The satellite will receive (uplink frequency) in the 14 GHz band and will transmit in the 12 GHz band.[9]

The United States and Canada will share time for direct broadcast experiments. Canada's main control station will be in Ottawa, using a 9-meter antenna. Some 20 terminals of three different types have been ordered for loan to experimenters in Canada:

(a) Two 10-foot terminals, costing approximately 0.5 million dollars each for development and production, will send and receive TV with two-way voice capability. Being transportable, these terminals can originate programs on location and will provide data/computer links testing the utility of such service, *e.g.*, for government offices.

(b) Seven or eight 7-foot terminals costing somewhat less than $200,000 each will be provided. These terminals will have a TV receive-only capability but will provide two-way voice, data, and sound services.

(c) Seven to ten 3-foot terminals costing about $70,000 each will also be provided. These terminals will be essentially two-way voice terminals with data capability and with

[8]B. Blevis & N. Davies, *supra* note 2; Communications Canada, *supra* note 2.

[9]Hearings, *supra* note 1, at 676-78.

the capacity to provide interconnections through the main or remote terminals. They will also be able to receive TV under ideal conditions, but in somewhat degraded form. The two-way voice capability is considered especially important because telephone service is absent through very large areas of northern Canada where microwave links are prevented by the terrain.

Additional receivers can also be brought into the program. Receivers equipped to receive only video will cost $5,000 to $6,000. Those equipped to receive video and to permit a voice response will cost $12,000 to $14,000. Receivers which are fully interactive (two-way TV and voice) will cost an estimated $50,000 after development costs have been met.[10]

The fundamental objectives of the Canadian program are to test various broadcast modes and services to remote areas, both in technical and programmatic terms. Approximately 40 experiments, twenty in the social field and twenty in the technical field are to be conducted. Both one-way and interactive modes will be tested. The experiments are in such fields as telemedicine, teleeducation (as for Eskimo and Indian groups), curriculum exchange, community development (including the interconnection of communities), and provincial government management and technology, including local access to data communication and teletype.[11] The particular experiments will be selected by a three-person committee which will forward its recommendations to the Minister of Communications. The committee will include a physicist, a sociologist, and an economist.

The Canadian case is particularly interesting. Here, for the first time, two nations—the United States and Canada—will collaborate in sharing the use of a direct broadcast facility. Either will in principle have the ability to broadcast programs into the other's receivers, but it may safely be assumed that this will not be done except by prearrangement; both nations have primarily domestic objectives.

IV.

The Japanese project is again a domestic broadcast experiment. The satellite is being procured by Japan through a U.S. manufacturer and will be launched by NASA in 1977 on a reimbursable basis. Japan plans to broadcast with a 100-watt power level, providing two color TV channels plus voice channels. The satellite will be located on the equator, approximately over Borneo. It will keep station with an accuracy of plus or minus 0.1 degree and a pointing capability of plus or minus 0.2 degree. The uplink will be in the 14 GHz band and the downlink will be in the 12 GHz band, again essentially avoiding interference with terrestrial services. The antenna beam will be focused as narrowly as possible on the Japanese Islands to minimize radiation impinging on mainland China, Korea, and Siberia, even though there is no basis for believing that radiation

[10] B. Blevis & N. Davies, *supra* note 2; Communications Canada, *supra* note 2.

[11] *Id.*

spillover could cause any difficulty or interference with terrestrial users because of the frequencies involved. Those regions could receive Japanese programs only if they wished to do so and consequently procured or duplicated Japanese receivers.[12]

The objectives of the Japanese program are to test standards, controls, techniques, and the effectiveness of direct broadcast modes, particularly in order to reach remote areas and to circumvent urban shadow.[13] Four types of terminals will be tested. The largest will be transportable stations having a 4.5-meter antenna and two-way TV and voice capability. Next will be a mobile station with a 2.5-meter antenna, also having two-way TV and voice capability. The other two terminals will be 2.5- to 4.5-meter-antenna receive-only stations and 1- to 1.6-meter-antenna high-quality community receivers with receive-only capability. The latter will cost $1,000-$3,000, and will require rigid antenna construction and very careful orientation for high-quality reception purposes.[14]

In summary, the broadcast activities in all three projects will not reach other countries' TV receivers unless those countries take deliberate steps to enable themselves to receive such broadcasts. In all three projects, entirely domestic purposes are intended.

What else do such projects suggest with regard to the issue of intrusive broadcasting? Imagine a time when several countries will be conducting operational direct/community broadcasting programs in the appropriate and duly assigned 12 and 14 GHz bands. Those bands afford adequate space for foreseeable national broadcasting programs to operate on the basis of assigned frequencies peculiar to each program within those bands. Each such use of those bands is permitted under ITU regulations on a noninterference basis only.[15] As additional countries enter the direct broadcast fraternity, they will be required to select broadcast and receive frequencies within those bands that will not interfere with countries already operating within those bands. Suppose a broadcasting nation wishes deliberately to transmit programs into another country's direct broadcast system. Broadcasting into channels already in use within the target country would be totally ineffective as the result would be interference rather than the clear reception of any program. Broadcasting into unoccupied channels is, of course, quite possible in theory, but it is reasonable to conclude that no nation will long make available unoccupied channels for unwelcome foreign broadcasts. Thus, in either situation the issue of intrusive broadcasting shrinks to inconsiderable proportions when viewed within the scope of realities of operational practice and technology.

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[12]National Space Development Agency (of Japan), Technical Specification of Japanese Broadcast Satellite (1974).

[13]*Id.*; Chiba, *supra* note 3.

[14]National Space Development Agency, *supra* note 12.

[15]*See* Partial Revision of the Radio Regulations, Geneva, 1971 and Final Protocol: Space Telecommunications, July 17, 1971, [1972] 23 U.S.T. 1527, T.I.A.S. No. 7435 (effective Jan. 1, 1973).

Given the objectives of the Indian, Canadian and Japanese direct/community broadcast experiments, and the impressive technical and programmatic competence available to these states, one can only conclude that the prospective benefits of direct broadcasting far outweigh the fears which have developed out of abstract discussion.